

NAME: _____

PERIOD: _____

DATE: _____

Which Pass Rate Answers Which Question?

AP Statistics · Topic 2.3 (CED 2.2) · B: 2026-10-16 · E: 2026-10-30

[STAR] TODAY'S PROBLEM

Suppose a school records exam outcomes for 300 students. Students chose for themselves whether to use the tutoring center.

Mara: *The pass rate is 75%.*

Noah: *The pass rate is 88%.*

Elena: *Tutoring raises the pass rate by 13 percentage points because $88\% - 75\% = 13$ percentage points.*

Exam outcomes for 300 students

Center?	Passed	Did not pass	Total
Used	66	9	75
Did not	159	66	225
Total	225	75	300

FIRST TAKE — before the video (5 min)

Can 75% and 88% both be correct pass rates? State what group you think each percentage describes.

[BOOK] NAME THE DENOMINATOR, THEN THE CLAIM (keep on the page)

A **marginal relative frequency** uses a row or column total divided by the table total. A **conditional relative frequency** uses a cell count divided by the total for the named group. To decide whether two categorical variables are associated, compare the same response's conditional percentage across the groups. Different conditional distributions show association. They do not show cause when people choose their own groups; a fair causal comparison requires random assignment.

Finish after the video:

DOK 1 (a) Identify 75% and 88% as marginal or conditional relative frequencies. For the conditional frequency, name the group on which it is conditioned.

DOK 2 (b) Verify both percentages from the table. Then compute the pass rate among students who did not use the center and the percentage-point gap between the two user groups.

DOK 3 (c) Explain which question Mara's 75% answers and which question Noah's 88% answers. Decide whether tutoring use and exam outcome appear associated, citing the two conditional pass rates that settle it. Then evaluate Elena's 13-point claim: state what two rates she compared, what comparison directly measures the group gap, and why the word *raises* is not warranted when students chose tutoring.

Frame: Mara answers _____, while Noah answers _____.

Frame: The variables appear _____ because _____; the word *raises* is unwarranted because _____.

EXIT — turn this in

One thing the video changed about my first take: _____